

Foundations of Deep Learning—Written Exam

Master's Degree in AI & Society (PSL-PSAI, Spring 2026)

Exam 0

Section	Topic	Suggested Time	Points
1	True / False	15 min	16
2	Multiple Choice	20 min	24
3	Short Answer	25 min	30
4	Exercises	30 min	30
Total		90 min	100

Part 1—True / False (16 points)

Instructions: Cross **True** or **False**. No explanation is required.

Scoring: +2 correct, -1 incorrect, 0 blank.

1.1 A multilayer perceptron in which every hidden layer uses a linear activation function is strictly equivalent to a single affine transformation, regardless of the number of hidden layers or their widths.

☐ True ☐ False

1.2 For a regression task, minimising the sum of squared errors (SSE) loss is equivalent to maximum likelihood estimation under the assumption that observations are corrupted by additive, zero-mean Gaussian noise with constant variance.

$$\mathcal{L}^{\text{SSE}}(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N (y_n(\mathbf{x}_n, \mathbf{w}) - t_n)^2$$

☐ True ☐ False

1.3 Gradient descent is guaranteed to converge to the global minimum of the loss function if the learning rate is chosen small enough.

☐ True ☐ False

1.4 Consider the softmax function:

$$\text{softmax}(a_i) = \frac{e^{a_i}}{\sum_j e^{a_j}}$$

The softmax output is unchanged when all logits are shifted by a constant α (i.e. $a_i \leftarrow a_i + \delta$), and also when all logits are scaled by a positive constant $c \neq 1$ (i.e. $a_i \leftarrow c \cdot a_i$).

☐ True ☐ False

1.5 Batch normalisation behaves identically during training and inference: in both cases it normalises each feature using the mean and variance of the current input batch.

☐ True ☐ False

1.6 In a convolutional layer with C_{out} filters of spatial size $M \times M$ applied to a C_{in} channel input of spatial dimensions $H \times W$, the number of trainable parameters is $(C_{in}M^2 + 1)C_{out}$. For any practical input size ($H, W \gg M$) this is fewer parameters than the equivalent fully connected layer mapping the same input to C_{out} output feature maps of the same spatial size.

☐ True ☐ False

1.7 He initialisation sets the variance of the weights of each layer l to $2/M^{(l-1)}$, and is specifically designed for networks using ReLU activation functions.

☐ True ☐ False

1.8 A transformer encoder trained without positional encodings would assign the same contextual representation to the sentence “*The cat chased the dog*” and the sentence “*The dog chased the cat*”, since both contain the same tokens.

☐ True ☐ False

Part 2—Multiple Choice (24 points)

Instructions: Each question states the number of correct answers k . You **must tick exactly k boxes**. No explanation is required.

Scoring: 4 points if all k ticked options are correct. 2 points if exactly $k - 1$ ticked options are correct. 0 otherwise. Furthermore, any answer with a number of ticked boxes different from k scores 0.

2.1 Consider a K -class classification problem. The network outputs logits $\mathbf{a} \in \mathbb{R}^K$ fed into a softmax, followed by cross-entropy loss. Which of the following are correct? (**2 correct answers**)

- ☐ (A) The gradient of the cross-entropy loss with respect to the logits is $\mathbf{y} - \mathbf{t}$, where $\mathbf{y}$ is the softmax output and $\mathbf{t}$ is the one-hot target.
- ☐ (B) The cross-entropy loss can reach exactly zero if the logits are chosen appropriately.
- ☐ (C) Minimising the cross-entropy loss is equivalent to maximising the log-likelihood under a Gaussian distribution.
- ☐ (D) For binary classification ($K = 2$), softmax + cross-entropy is mathematically equivalent to a sigmoid output + binary cross-entropy.
- ☐ (E) Replacing softmax with independent sigmoid activations on each output unit produces an equivalent model for single-label K -class classification.

2.2 Which of the following statements about the geometry of loss landscapes and gradient descent are correct? (**3 correct answers**)

- ☐ (A) In high-dimensional parameter spaces, saddle points are exceedingly rare compared to local minima.
- ☐ (B) A saddle point is a stationary point that is a local minimum along some directions and a local maximum along others.
- ☐ (C) In a loss landscape with stiff and soft directions, SGD with a fixed learning rate faces a dilemma: a learning rate large enough to make progress in soft directions will cause oscillations or divergence in stiff directions.
- ☐ (D) Adaptive optimisers such as RMSProp and Adam mitigate the stiff/soft direction problem by rescaling the gradient in each parameter direction according to the local curvature.
- ☐ (E) Momentum and Adam both overcome the stiff/soft direction problem by rescaling the gradient component-wise based on past gradient magnitudes.

2.3 Which of the following statements about backpropagation and automatic differentiation are correct? (**3 correct answers**)

- ☐ (A) Backpropagation computes the gradient of the loss with respect to all parameters in a single backward pass, at a computational cost comparable to one forward pass.
- ☐ (B) Reverse-mode automatic differentiation requires the computational graph to be a directed acyclic graph (DAG).
- ☐ (C) The gradient of the loss with respect to a weight in an early layer depends only on quantities local to that layer, independently of the weights and activations in later layers.
- ☐ (D) The error signal $\delta_j^{(l)}$ at layer l is independent of the activation function used at that layer, since the activation function only affects the forward pass.

☐ (E) During the backward pass, the intermediate activations $z_j^{(l)}$ (or pre-activations $a_j^{(l)}$) computed during the forward pass must be stored in memory, as they are required to evaluate the error signals $\delta_j^{(l)}$.

2.4 Which of the following statements about regularisation are correct? (**2 correct answers**)

- ☐ (A) Increasing the number of parameters beyond the point where a model perfectly fits the training data always leads to higher test error.
- ☐ (B) Dropout prevents neurons from co-adapting by randomly disabling units during training, forcing each unit to learn representations that are useful independently rather than in conjunction with specific other units.
- ☐ (C) Dropout should be applied both during training and at inference time to ensure consistency between the two phases.
- ☐ (D) By constraining the hypothesis space, the inductive biases introduced by regularisation techniques are guaranteed to improve generalisation.
- ☐ (E) L2 regularisation penalises large weights and biases the optimiser towards solutions with smaller weight magnitudes.
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2.5 Which of the following statements about convolutional neural networks are correct? (**3 correct answers**)

- ☐ (A) The receptive field of a unit in a deep CNN grows with the depth of the network, allowing deeper layers to capture larger-scale patterns than shallower layers.
- ☐ (B) A max pooling layer and a convolutional layer with the same window size, stride, and padding have the same number of learnable parameters.
- ☐ (C) A 1×1 convolution can change the number of channels without altering the spatial dimensions of the feature map.
- ☐ (D) Residual (skip) connections help mitigate the vanishing gradient problem by providing an additive shortcut path through which gradients can flow directly.
- ☐ (E) A standard CNN is translation-invariant: shifting the input image always produces identical classification scores.
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2.6 Which of the following statements about the transformer architecture are correct? (**2 correct answers**)

- ☐ (A) The memory required to store the attention matrix in self-attention scales linearly with the sequence length N .
- ☐ (B) Using separate learnable projection matrices $W^{(q)}$, $W^{(k)}$, $W^{(v)}$ for queries, keys and values is necessary to allow asymmetric attention: using a single shared projection for both queries and keys would force the attention weights to be symmetric.

- ☐ (C) In multi-head attention with H heads and model (embedding) dimension D , each head projects the input to dimension $D_k = D_v = D/H$, so the total number of parameters across all heads is H times larger than the ones for a single attention head with dimension D .
 - ☐ (D) Transformers use layer normalisation rather than batch normalisation because layer normalisation is computationally cheaper: it normalises over the feature dimension rather than the batch dimension, reducing the number of operations.
 - ☐ (E) The $1/\sqrt{D_k}$ scaling in scaled dot-product attention is needed because, without it, the dot products between queries and keys grow in magnitude with D_k , causing gradients to vanish.
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Part 3 — Short Answer (30 points)

Instructions: Answer concisely. Use equations if/where appropriate. **Aim for 5–10 sentences per sub-question.**

3.1 Vanishing gradients (15 points)

- (a) Explain the vanishing gradient problem and why it makes training deep networks difficult. Your answer should reference the chain rule and the mathematical mechanism by which gradients shrink. (5 points)
 - (b) Why is the problem particularly severe with sigmoid or tanh activations? Provide a quantitative argument using the derivative of the sigmoid. (4 points)
 - (c) Describe residual/skip connections as a strategy to mitigate the vanishing gradients problem. (6 points)
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3.2 Natural vs. Artificial Intelligence (15 points)

- (a) Biological brains learn continuously throughout a lifetime without forgetting past knowledge. Standard neural networks trained with gradient descent struggle with this. Describe this limitation and explain why it occurs. Then describe the paradigm of *continual learning* as a response to this challenge, and name one of the approaches it proposes. (5 points)
 - (b) What is transfer learning? Describe the two main strategies discussed in the course and give one concrete example seen in one of the lectures or group-presentations. Explain why features learned on a source task transfer to a different target task. (5 points)
 - (c) In what sense do AI systems trained on text or images possess *statistics about the world* but not *experience of it*? Describe briefly one research direction that attempts to close this gap. (5 points)
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Part 4 — Exercises (30 points)

4.1 CNN Architecture (15 points)

Consider the following LeNet-like architecture (LeCun et al., 1998) for classifying 28×28 greyscale images of handwritten digits into 10 classes. All convolutional and fully connected layers include biases.

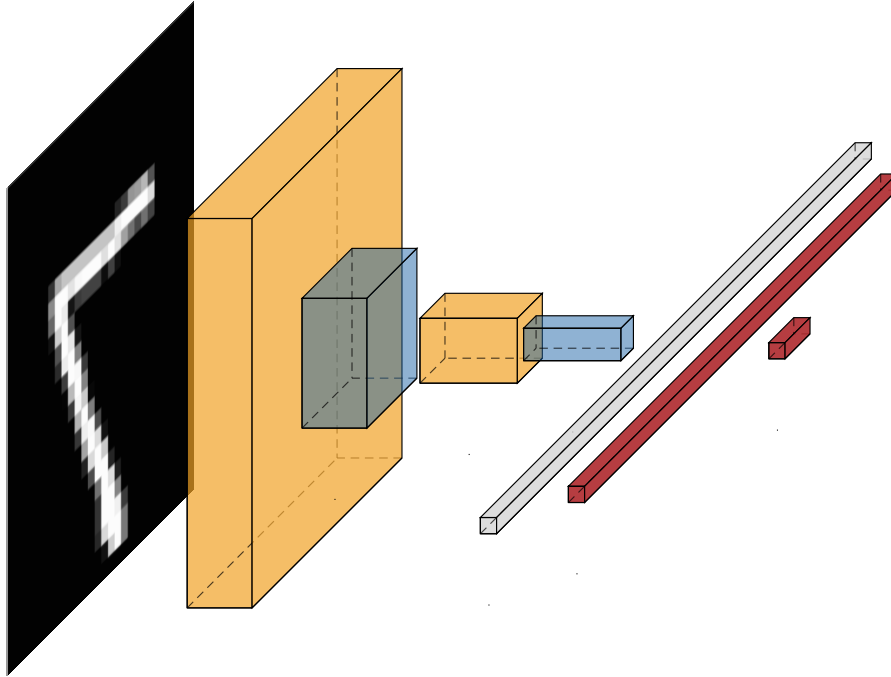


Figure 1: Computational graph for Exercise 4.1

Layer	Description
INPUT	$28 \times 28 \times 1$
CONV-1	32 filters, 5×5 kernel, no padding, stride 1
POOL-1	Max pooling, 3×3 , stride 3
CONV-2	64 filters, 5×5 kernel, no padding, stride 1
POOL-2	Max pooling, 2×2 , stride 2
FLATTEN	—
FC-1	200 neurons
FC-2	10 neurons (output)

(a) (8 pts) Complete the table below. Write output dimensions as (H, W, C) for volumetric layers and $(N,)$ for flat layers.

Layer	Output dimensions	# weight params	# bias params
INPUT	(28, 28, 1)	—	—
CONV-1			
POOL-1			
CONV-2			
POOL-2			
FLATTEN			
FC-1			
FC-2			

(b) (4 pts) What is the total number of trainable parameters? Which layer(s) contribute the most, and what does this reveal about the architecture?

(c) (3 pts) In the original LeNet, the input images are 28×28 pixels. If the input were instead 32×32 pixels (all other layers unchanged), what would the FLATTEN dimension be? How many weight parameters would FC-1 have?

4.2 Automatic Differentiation (15 points)

Consider a minimal neural network with scalar input $x \in \mathbb{R}$, learnable parameters $w \in \mathbb{R}$, $b \in \mathbb{R}$, target $t \in \mathbb{R}$, and loss:

$$y = \text{ReLU}(wx + b), \quad \mathcal{L} = (y - t)^2$$

The computational graph is given below. Each node v_i corresponds to a quantity in the network and $v_9 = \mathcal{L}$.

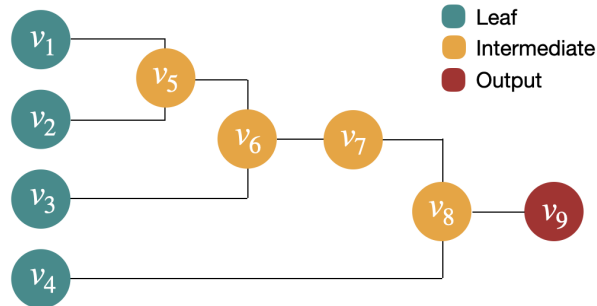


Figure 2: Computational graph for Exercise 4.2

The values for a particular training example are: $x = 2$, $w = 3$, $b = -7$, $t = 1$.

(a) (4 pts) Specify the correspondence between each node and the quantity it represents. Specifically, for leaf nodes, identify the value they correspond to (e.g. $v_1 = x$). The expressions for the intermediate and output nodes should also be written in terms of the nodes they depend on.

(b) (4 pts) Carry out the **forward pass**. Compute the value of each node v_1 through v_9 for the given inputs.

(c) (7 pts) Carry out the **backward pass**. Compute the following gradients step by step, stating the local gradient at each node:

$$\frac{\partial \mathcal{L}}{\partial (wx + b)}, \quad \frac{\partial \mathcal{L}}{\partial b}, \quad \frac{\partial \mathcal{L}}{\partial w}.$$

What do you observe about the gradients with respect to w and b ? Connect your answer to a phenomenon discussed in the course and explain its practical consequence for training.

Useful Formulas

Activation Functions

$$\sigma(a) = \frac{1}{1 + e^{-a}}, \quad \sigma'(a) = \sigma(a)(1 - \sigma(a))$$

$$\tanh(a) = \frac{e^a - e^{-a}}{e^a + e^{-a}}, \quad \tanh'(a) = 1 - \tanh^2(a)$$

$$\text{ReLU}(a) = \max(0, a), \quad \text{ReLU}'(a) = \begin{cases} 1 & \text{if } a > 0 \\ 0 & \text{if } a < 0 \end{cases}$$

$$\text{softmax}(a_i) = \frac{e^{a_i}}{\sum_j e^{a_j}}$$

Convolutional Layer — Output Spatial Size

For a convolutional (or pooling) layer with input size H_{in} , kernel size M , padding p , and stride S :

$$H_{out} = \left\lfloor \frac{H_{in} + 2p - M}{S} \right\rfloor + 1$$

Automatic Differentiation—Adjoint Variables

For a computational graph with nodes $v_1, \dots, v_n$ and loss $\mathcal{L} = v_n$, the adjoint of node v_i is $\bar{v}_i \equiv \partial \mathcal{L} / \partial v_i$.

Reverse recursion (from output to inputs):

$$\bar{v}_i = \sum_{j \in \text{ch}(i)} \bar{v}_j \frac{\partial v_j}{\partial v_i}$$

where $\text{ch}(i)$ denotes the set of children of node v_i in the graph.